Predicting Coronary Heart Disease from Survey Data: What Helped Our Linear Models?

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Abstract—We compared six linear methods and two support vector machine (SVM) losses for predicting self-reported coronary heart disease from the BRFSS survey. We first cleaned survey answers and built a shared set of features. Logistic regression was the strongest initial model, with local test F1 0.42456. We then investigated changes to the SVM. Squaring its hinge loss improved F1 from 0.22464 to 0.42332. Giving positive examples more weight and keeping more questions with missing answers further increased F1 to 0.42672. Adding age and BMI combinations did not improve validation F1 enough to keep them. All model choices used validation predictions within the training portion of an 80/20 split. Reported test scores are local, not competition leaderboard scores.

I. Task and Experimental Setup

We predict self-reported coronary heart disease (MICHD), including heart attack or angina [1]. The data contains 328,135 labeled respondents, 109,379 competition respondents and 321 input fields. Only 8.83% of labeled respondents are positive. Predicting "no disease" for everyone would give 91.17% accuracy while finding none of the positive cases. We therefore optimize F1, which balances precision (how many predicted positives are correct) and recall (how many actual positives are found).

How we compared models. We used a stratified 80/20 split (seed 42), preserving the class proportions: 262,508 development rows for training and model selection, and 65,627 rows for local testing. Within development, we used two-fold cross-validation (seed 104). Training on each half and predicting the other gives one out-of-fold (OOF) score per development respondent.

Every model produces a score. We chose the cutoff for predicting disease by maximizing F1 on the combined OOF scores, checking every distinct score as a possible threshold. We then trained on all development rows and used this fixed threshold on the local test set. All preprocessing statistics were learned only from the training rows of each fold. Exported competition predictions use the same models trained on 80% of the labeled data.

II. Preparing the Survey Data

We used the same cleaning steps for all initial models. Survey answer codes need special care: a large number may mean "refused" rather than an actual measurement. We used field-specific rules from the 2015 codebook [2]. For example, employment code 7 means "retired" and stays valid, while health-day code 88 means zero days. Unknown/refused answers become missing values. For fields mixing units, we use calculated versions already in the dataset, such as uniformly measured weight.

We removed administrative identifiers, dates and sampling weights. We also excluded HAREHAB1, a rehabilitation question asked after a heart attack, because it could reveal the target. State remained a categorical feature. Initially, we removed questions with more than 80% missing answers.

For numeric fields, we filled missing answers with the training median and standardized values using the training mean and standard deviation. We also added binary indicators showing which answers were missing. For categorical fields, we used one-hot encoding: separate binary columns represent the possible answers, with one category omitted as a reference. Missing answers have their own indicator. New categories unseen during training use the reference encoding.

Finally, we added an intercept and removed constant or redundant columns to make least squares solvable. This pipeline, in data-preprocessing.py, kept 174 fields and produced 559 features. We did not measure each cleaning step's individual contribution.

III. Experiments and Results

A. Which initial model worked best?

We tried the six course methods: least squares, ridge regression, MSE regression trained by gradient descent (GD) or single-example stochastic GD (SGD), logistic regression, and regularized logistic regression. MSE methods fit numeric targets $-1/+1$; logistic methods fit 0/1 labels. Ridge and regularized logistic add an L_2 penalty, discouraging large weights, with strengths $\lambda = 10^{-4}$ and 10^{-3} respectively. We also tried an ordinary hinge-loss SVM with $\lambda = 10^{-3}$.

Logistic regression was best (Table I). The four MSE methods gave similar F1 scores. Regularization did not improve least squares or logistic at the tested strengths. All initial fits met their stopping rules. The ordinary SVM performed poorly, motivating the next experiments.

B. Did changing the SVM loss help?

Ordinary hinge loss penalizes mistakes and predictions too close to the decision boundary. We tried half its square to emphasize larger errors. Both use penalty $\lambda\|w\|^2/2$, including the intercept, with $\lambda = 10^{-3}$.

Table I
Initial model comparison and the squared-hinge alternative. All use the same 559-feature pipeline. Test means the local 20% split.

Method	OOF F1	Test F1	Test accuracy
MSE GD	0.41935	0.41639	0.85780
MSE SGD	0.41983	0.41778	0.87267
Least squares	0.42036	0.41759	0.86157
Ridge, $\lambda = 10^{-4}$	0.41891	0.41706	0.85965
Logistic	0.42652	0.42456	0.86947
Reg. logistic, $\lambda = 10^{-3}$	0.42315	0.41880	0.86492
Hinge SVM, $\lambda = 10^{-3}$	0.25706	0.22464	0.75344
Squared-hinge SVM, 10^{-3}	0.42637	0.42332	0.86266

Table II
Squared-hinge experiments. Baseline: $\lambda = 10^{-3}$, positive weight 1, 80% missingness cutoff, no added age/BMI terms. Each model has its own OOF-selected threshold.

Setting	OOF F1	Test F1
$\lambda = 10^{-5}$ (not converged)	0.41592	0.41363
$\lambda = 10^{-4}$	0.42583	0.42463
Baseline, $\lambda = 10^{-3}$	0.42637	0.42332
$\lambda = 10^{-2}$	0.42462	0.42386
Baseline + three age/BMI terms	0.42656	0.42381
Baseline + positive weight 2	0.42801	0.42345
Baseline + positive weight 4	0.42836	0.42475
Weight 4 + 95% cutoff (selected)	0.42966	0.42672

This change helped substantially: test F1 increased from 0.22464 to 0.42332, close to logistic regression. Diagnostics on 10,000 labeled rows showed that ordinary-SVM scores were concentrated near -1, with standard deviation 0.00634. Its selected threshold was also near -1, making predictions sensitive to small score differences. Meeting the stopping rule did not guarantee a useful classifier.

C. Did changing regularization or adding features help?

Starting from squared hinge, we tested four regularization strengths (Table II). We selected $\lambda = 10^{-3}$, which gave the best OOF F1. The 10^{-5} run did not finish converging within its update budget and was excluded from selection. Its score does not represent a fully trained model.

We next added three features: squared standardized age, squared standardized BMI, and their product. These let a linear model express some curved or combined effects of age and BMI. OOF F1 increased by only 0.00019, below our 0.001 minimum improvement, so we did not keep them.

D. Did emphasizing positive cases and keeping sparse fields help?

To address the small positive class, we multiplied positive examples' squared-hinge loss by 2 or 4 relative to negatives. We normalized sample weights to have mean one, keeping the overall loss scale comparable. Weight 4 gave the best OOF F1, improving from 0.42637 to 0.42836. Thus, class weighting helped even though the threshold was already tuned.

Keeping weight 4, we then allowed questions with up to 95% missing answers, instead of 80%. This retained 221 fields and produced 713 features. OOF F1 rose again, to 0.42966, suggesting that some sparse questions still contained useful information. Each accepted change improved both folds and gained at least 0.001 pooled OOF F1. The selected model reached test F1 0.42672, precision 0.34424 and recall 0.56117: it found about 56% of positive cases, while about 34% of its positive predictions were correct.

IV. Conclusions and Reproducibility

Changing the SVM loss gave the largest improvement in these experiments. Class weighting and retaining more sparse fields added smaller gains. The tested age/BMI features did not help enough to keep, and the initial regularized regression models did not outperform ordinary logistic. Our selected weighted squared-hinge SVM improved test F1 by 0.00340 over its unweighted baseline and 0.00216 over logistic regression. These are small gains on one split; we did not estimate uncertainty across repeated splits. The local test set had been inspected in earlier experiments, although it was excluded from selection in this staged search.

For reproducibility, all iterative models start at zero. Batch GD stops at maximum absolute gradient 10^{-4} (cap: one million updates). SGD stops when its training half-MSE is within 10^{-4} of exact least squares at two consecutive checks (cap: 60 million updates). Automatic step sizes and whitening (rescaling feature combinations) help optimization; whitening is used for SGD and ordinary logistic. SVMs use dual coordinate ascent, updating one example at a time, until a bound on remaining optimization error is at most 10^{-4}. The cap is five million updates for the initial comparison and twenty million for later experiments. The selected model needed 8.3 million. Only converged models were eligible for tuning; we kept the default λ if within 0.001 OOF F1 of the best candidate.

Code uses NumPy and the standard library; result paths are listed in the source comments. No verified leaderboard score is saved for this model. Reproduce its settings with:

```
python run.py --model svm_squared_hinge \
  --lambda 0.001 --positive-weight 4 \
  --max-missing 0.95 --svm-iters 20000000 \
  --export --output-dir output/report_reproduction
```

References

[1] EPFL CS-433, "Project 1: Classification," Fall 2026, provided project description and BRFSS dataset.

[2] Centers for Disease Control and Prevention, "2015 BRFSS Codebook," https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf.